$$D_{\mathrm{Hb}}(\tau) = \frac{\min_{a,b} \sum_j [\ell_j - r_j - a\int \phi_j dt - b\int \phi_j e^{-(t-t_0)/\tau} dt]^2}{\sum_j [(\int \phi_j \bar{R} dt)^2 + \rho^2 (\int \phi_j \bar{H} dt)^2]}$$